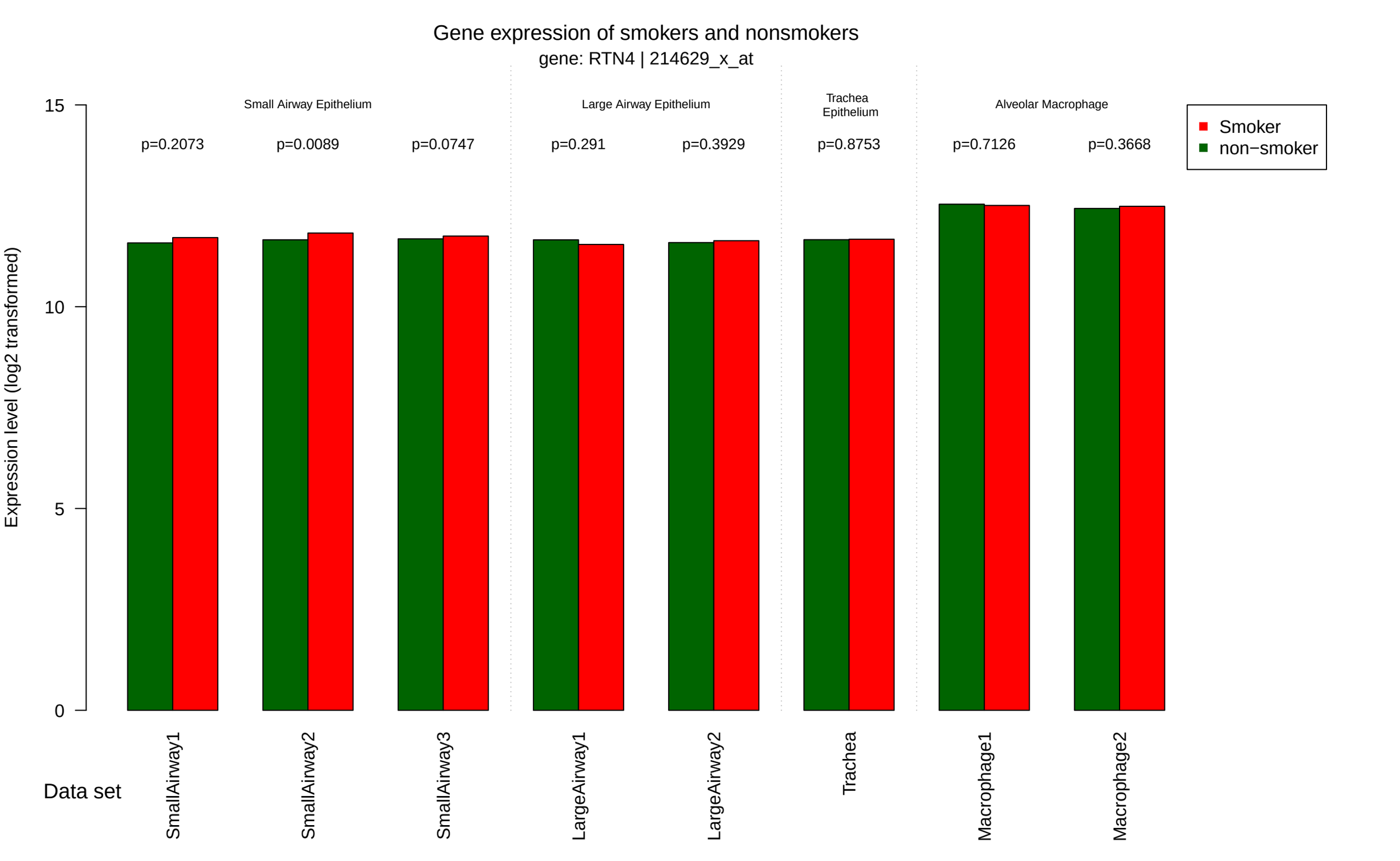
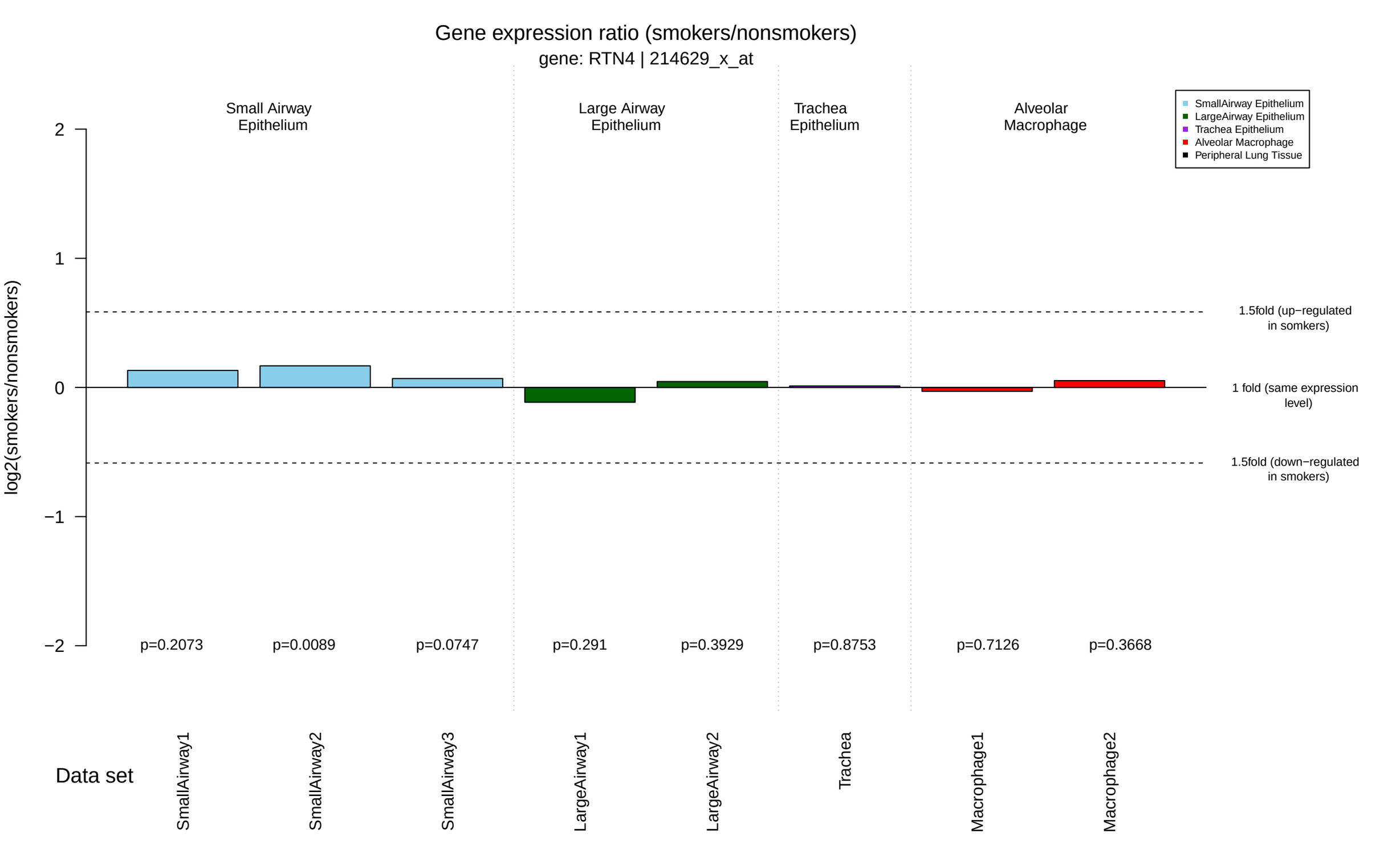
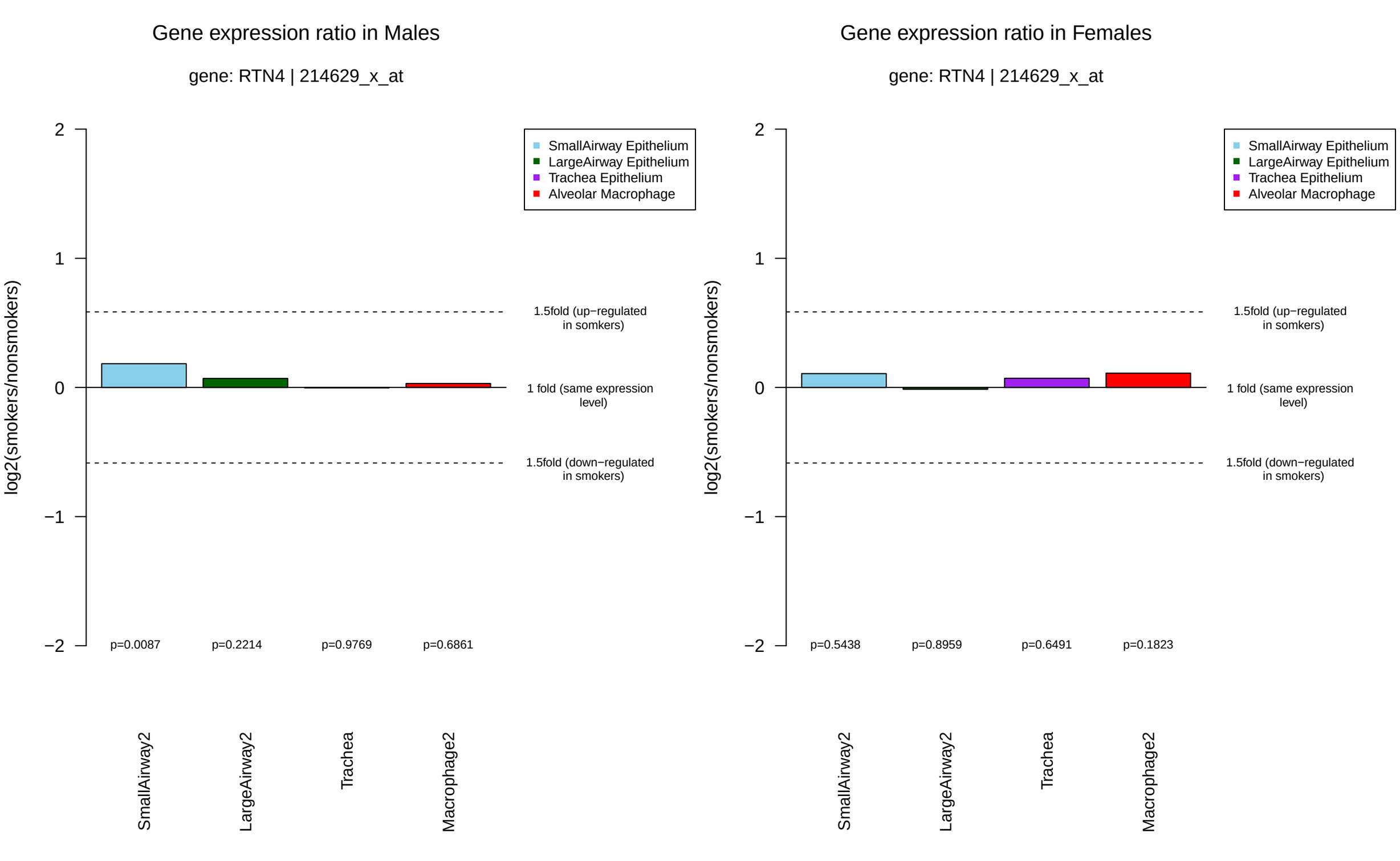
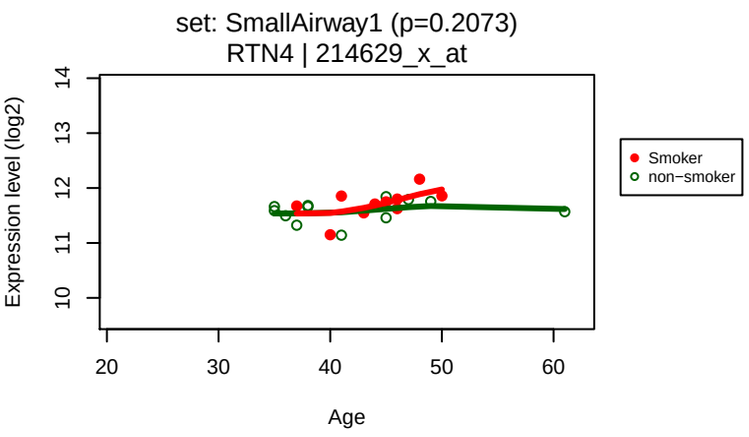
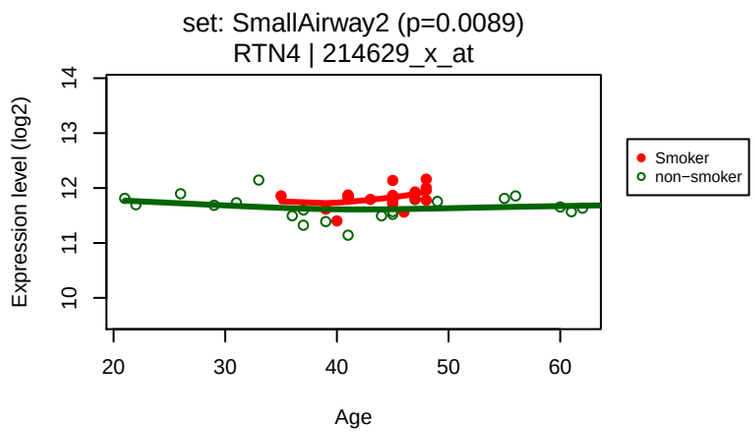
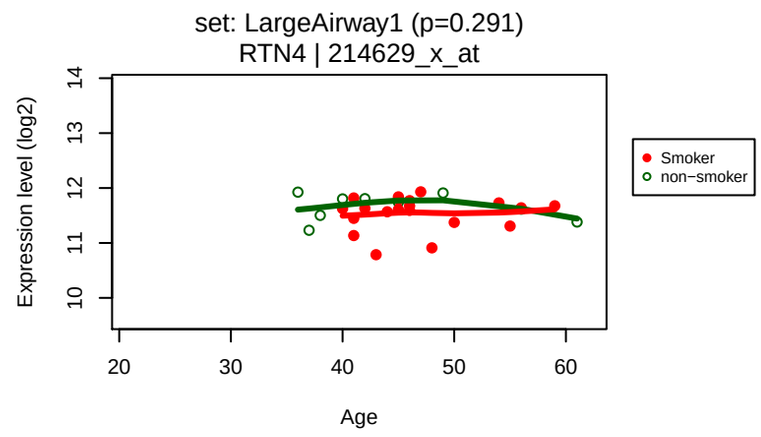

Gene expression of smokers and nonsmokers
gene: RTN4 | 214629_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2073
p=0.0089
p=0.0747
p=0.291
p=0.3929
p=0.8753
p=0.7126
p=0.3668
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

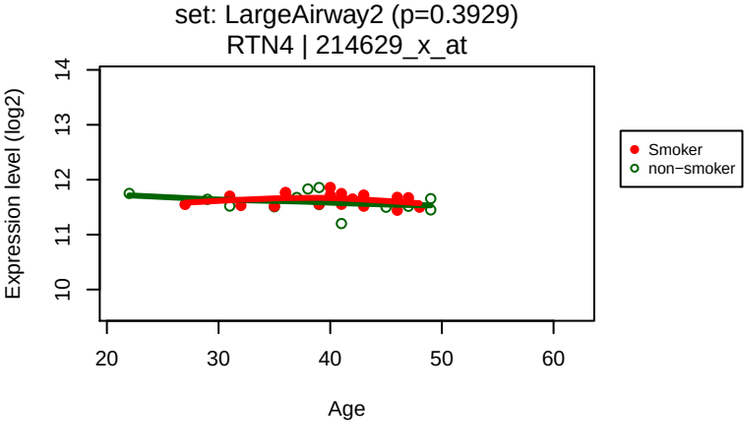

Gene expression ratio (smokers/nonsmokers)
gene: RTN4 | 214629_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2073
p=0.0089
p=0.0747
p=0.291
p=0.3929
p=0.8753
p=0.7126
p=0.3668
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

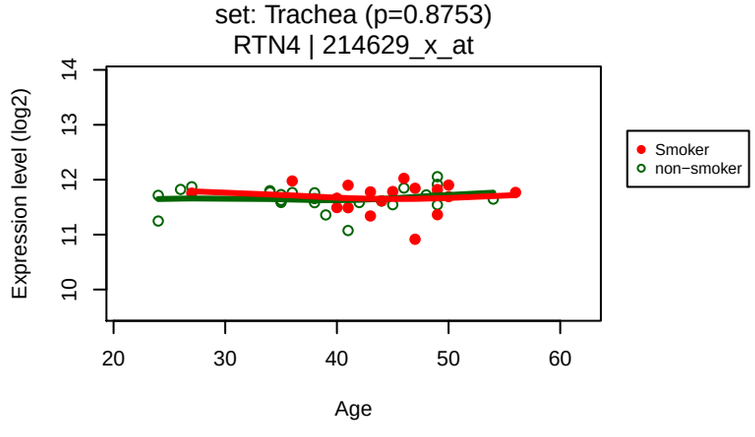

Gene expression ratio in Males
gene: RTN4 | 214629_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0087
p=0.2214
p=0.9769
p=0.6861
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RTN4 | 214629_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5438
p=0.8959
p=0.6491
p=0.1823
SmallAirway2
LargeAirway2
Trachea
Macrophage2

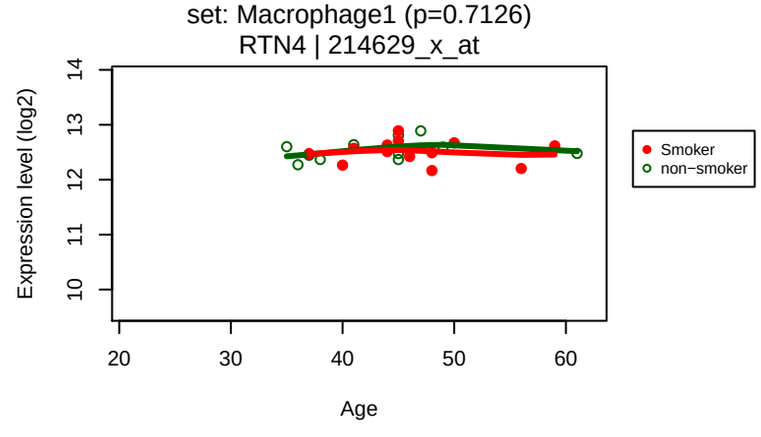

set: SmallAirway1 (p=0.2073)
RTN4 | 214629_x_at
Expression level (log2)
Age
Smoker
non−smoker

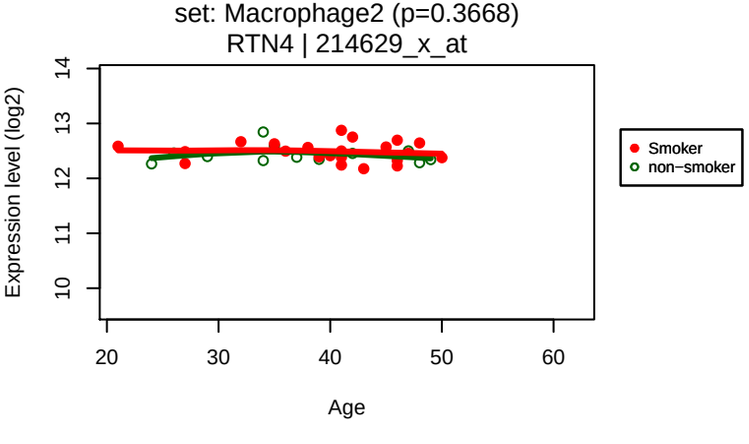

set: SmallAirway2 (p=0.0089)
RTN4 | 214629_x_at
Expression level (log2)
Age
Smoker
non−smoker

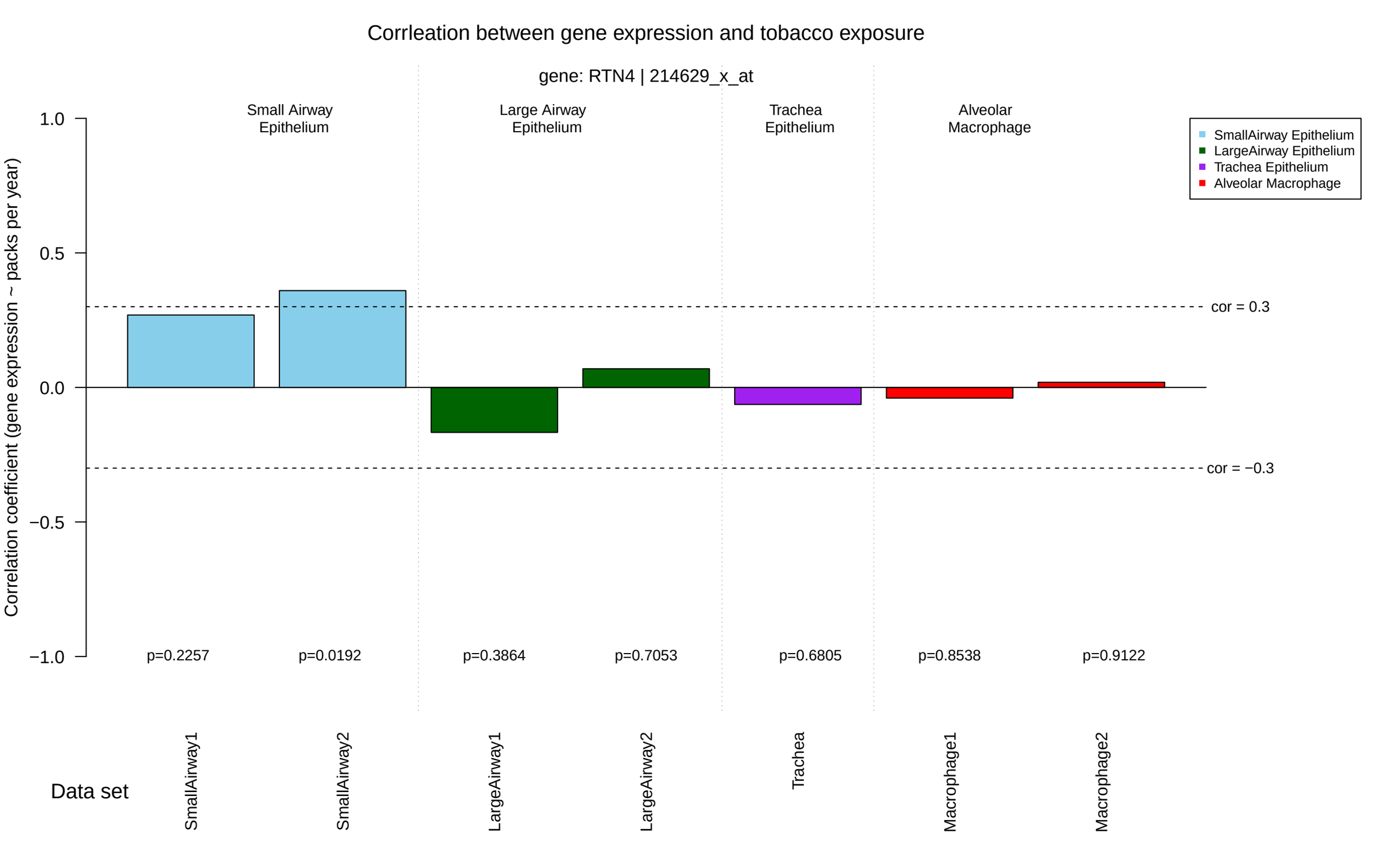

set: LargeAirway1 (p=0.291)
RTN4 | 214629_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3929)
RTN4 | 214629_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8753)
RTN4 | 214629_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7126)
RTN4 | 214629_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3668)
RTN4 | 214629_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RTN4 | 214629_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2257
p=0.0192
p=0.3864
p=0.7053
p=0.6805
p=0.8538
p=0.9122
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2